

# Question Bank — Operating Systems

Module 2 companion | PYQ-style + Practice | exam-tagged, answers & explanations

## ■ How to use this bank

**Two sections: PYQ-style** (modelled on real IBPS/SSC/GATE/PSU/DSSSB questions, exam-tagged) and **Practice** (full syllabus coverage by sub-topic).

Every question has answer + explanation in the green box.

**Tag legend:** IBPS SO IT • Coal India MT • DSSSB TGT • All 3 exams • GATE-level • SSC/other PSU.

**For IBPS:** prioritise IBPS / All-3; GATE-tagged numericals (scheduling/Banker's/paging math) are Coal-India depth.

## SECTION A — PYQ-style (modelled on actual exam questions)

Tagged by the exam whose style/level each reflects.

### Basics, types & structure

**Q1**

All 3 exams

The primary purpose of an operating system is to:

- (a) Compile programs
- (b) Manage hardware & software resources
- (c) Design circuits
- (d) Encrypt data

**Answer: (b)**

The OS manages resources (CPU, memory, I/O) and provides an interface between hardware and user programs.

**Q2**

All 3 exams

SSC/other

The core part of an OS that always stays in memory is the:

- (a) Shell
- (b) Kernel
- (c) Compiler
- (d) Loader

**Answer: (b)**

The kernel is the always-resident core managing processes, memory and devices; the shell is the command interpreter.

**Q3**

All 3 exams

A program in execution is called a:

- (a) Thread
- (b) Process
- (c) Function
- (d) Module

**Answer: (b)**

A process = program in execution (active). A program on disk is passive; a thread is a unit within a process.

**Q4**

IBPS SO IT

All 3 exams

**Which OS type guarantees a response within a strict time deadline?**

- (a) Batch
- (b) Time-sharing
- (c) Real-time
- (d) Distributed

**Answer: (c)**

A Real-Time OS (RTOS) guarantees deadlines — critical for airbags, medical and control systems.

**Q5**

DSSSB TGT

All 3 exams

**In which OS type does the CPU switch between jobs so users get interactive response?**

- (a) Batch
- (b) Time-sharing
- (c) Real-time
- (d) Single-user

**Answer: (b)**

Time-sharing (multitasking) slices CPU time among users/processes for interactivity.

**Q6**

Coal India MT

GATE-level

**The switch from user mode to kernel mode occurs via a:**

- (a) Context switch
- (b) System call / trap
- (c) Page fault only
- (d) Compiler call

**Answer: (b)**

A system call (trap) transitions from user to kernel mode to request OS services.

## Processes & scheduling

**Q7**

All 3 exams

**The structure that stores all information about a process is the:**

- (a) TLB
- (b) PCB
- (c) MMU
- (d) ALU

**Answer: (b)**

The Process Control Block (PCB) holds process ID, state, PC, registers, memory and file info.

**Q8**

All 3 exams

SSC/other

**Saving one process's context and loading another's is called a:**

- (a) System call
- (b) Context switch
- (c) Page fault
- (d) Interrupt

**Answer: (b)**

A context switch saves/restores PCBs when the CPU moves between processes — pure overhead.

**Q9**

All 3 exams

**In which state is a process waiting only for the CPU?**

- (a) Running
- (b) Ready
- (c) Waiting
- (d) Terminated

**Answer: (b)**

A Ready process has everything except the CPU. Running = on CPU; Waiting = blocked on I/O.

**Q10**

IBPS SO IT

All 3 exams

**Which scheduling algorithm can cause the 'convoy effect'?**

- (a) SJF
- (b) Round Robin
- (c) FCFS
- (d) Priority

**Answer: (c)**

FCFS can make short jobs wait behind a long one (convoy effect).

**Q11**

IBPS SO IT

All 3 exams

**Which scheduling algorithm gives the minimum average waiting time?**

- (a) FCFS
- (b) SJF
- (c) Round Robin
- (d) Priority

**Answer: (b)**

Shortest Job First (SJF) is provably optimal for average waiting time (needs burst-time knowledge).

**Q12**

All 3 exams

**Round Robin scheduling is most associated with which quantity?**

- (a) Priority number
- (b) Time quantum
- (c) Burst prediction
- (d) Deadline

**Answer: (b)**

Round Robin gives each process a fixed time quantum cyclically — good for time-sharing.

**Q13**

All 3 exams

**Priority scheduling can lead to starvation, which is solved by:**

- (a) Paging
- (b) Aging
- (c) Swapping
- (d) Spooling

**Answer: (b)**

Aging gradually raises the priority of long-waiting processes, preventing starvation.

**Q14****GATE-level****Coal India MT**

Processes arrive at time 0 with bursts  $P_1=4$ ,  $P_2=3$ ,  $P_3=1$ . Average waiting time under non-preemptive SJF is:

- (a) 1.33
- (b) 1.67
- (c) 2.33
- (d) 3.0

**Answer: (b)**

SJF order by burst:  $P_3(1)$ ,  $P_2(3)$ ,  $P_1(4)$ . Waiting times = 0, 1, 4  $\rightarrow$  average =  $5/3 \approx 1.67$ . Use the Gantt-chart method: waiting = start time – arrival.

## Synchronization & deadlocks

**Q15****All 3 exams**

The section of code accessing shared resources is called the:

- (a) Kernel section
- (b) Critical section
- (c) Ready queue
- (d) Page frame

**Answer: (b)**

The critical section is where shared data is accessed; needs mutual exclusion.

**Q16****All 3 exams****SSC/other**

A binary semaphore is essentially a:

- (a) Counter of 100
- (b) Mutex (mutual-exclusion lock)
- (c) Scheduler
- (d) Page table

**Answer: (b)**

A binary semaphore (0/1) acts as a mutex ensuring only one process in the critical section.

**Q17****All 3 exams**

The two atomic operations on a semaphore are:

- (a) push/pop
- (b) wait(P) and signal(V)
- (c) lock/free
- (d) read/write

**Answer: (b)**

Semaphore operations: wait (P, decrement) and signal (V, increment), both atomic.

**Q18****IBPS SO IT****All 3 exams**

How many Coffman conditions must ALL hold for a deadlock?

- (a) 2
- (b) 3
- (c) 4
- (d) 5

**Answer: (c)**

All four: mutual exclusion, hold-and-wait, no preemption, circular wait.

**Q19**

All 3 exams

Which is NOT one of the four deadlock (Coffman) conditions?

- (a) Mutual exclusion
- (b) Hold and wait
- (c) Circular wait
- (d) Aging

**Answer: (d)**

Aging is an anti-starvation technique, not a deadlock condition.

**Q20**

Coal India MT

GATE-level

Banker's Algorithm is used for deadlock:

- (a) Prevention
- (b) Avoidance
- (c) Detection
- (d) Recovery

**Answer: (b)**

Banker's Algorithm is an avoidance scheme — it grants requests only if the system stays in a safe state.

**Q21**

All 3 exams

Breaking the 'circular wait' condition (e.g. ordering resources) is a form of deadlock:

- (a) Avoidance
- (b) Prevention
- (c) Detection
- (d) Recovery

**Answer: (b)**

Ensuring a condition can never hold is prevention; resource ordering breaks circular wait.

## Memory management

**Q22**

All 3 exams

Which suffers from external fragmentation?

- (a) Paging
- (b) Segmentation
- (c) TLB
- (d) Both none

**Answer: (b)**

Segmentation (variable-size blocks) causes external fragmentation; paging causes internal fragmentation.

**Q23**

All 3 exams

SSC/other

Paging primarily suffers from:

- (a) External fragmentation
- (b) Internal fragmentation
- (c) No fragmentation
- (d) Thrashing only

**Answer: (b)**

Fixed-size pages waste space in the last page = internal fragmentation.

**Q24****IBPS SO IT****All 3 exams**

**A reference to a page not currently in memory causes a:**

- (a) Context switch
- (b) Page fault
- (c) Deadlock
- (d) Cache hit

**Answer: (b)**

A page fault occurs; the OS fetches the page from disk (demand paging).

**Q25****All 3 exams**

**Excessive paging where the CPU spends more time swapping than working is:**

- (a) Fragmentation
- (b) Thrashing
- (c) Spooling
- (d) Starvation

**Answer: (b)**

Thrashing = so many page faults that little useful work happens.

**Q26****All 3 exams**

**Which page-replacement algorithm can suffer Belady's anomaly?**

- (a) LRU
- (b) Optimal
- (c) FIFO
- (d) MRU

**Answer: (c)**

FIFO can show Belady's anomaly (more frames → more faults); LRU and Optimal do not.

**Q27****GATE-level****Coal India MT**

**Reference string 1,2,3,4,1,2,5 with 3 frames under FIFO gives how many page faults?**

- (a) 5
- (b) 6
- (c) 7
- (d) 4

**Answer: (b)**

FIFO trace (3 frames): 1F,2F,3F,4F(evict1),1F(evict2),2F(evict3),5F(evict4) = 6 faults. Every reference here misses because the evicted page is needed again soon.

**Q28****All 3 exams**

**The hardware that translates logical to physical addresses is the:**

- (a) ALU
- (b) MMU
- (c) TLB
- (d) PCB

**Answer: (b)**

The Memory Management Unit (MMU) maps logical (virtual) to physical addresses.

## File & disk

**Q29**

All 3 exams

**Which file allocation method allows fast random access with no external fragmentation?**

- (a) Contiguous
- (b) Linked
- (c) Indexed
- (d) Sequential

**Answer: (c)**

Indexed allocation keeps an index block of pointers, enabling random access without external fragmentation.

**Q30**

All 3 exams

SSC/other

**Which disk-scheduling algorithm serves the request closest to the current head position?**

- (a) FCFS
- (b) SSTF
- (c) SCAN
- (d) C-SCAN

**Answer: (b)**

SSTF (Shortest Seek Time First) picks the nearest request; risks starvation of far requests.

**Q31**

All 3 exams

**The SCAN disk-scheduling algorithm is also called the:**

- (a) Circular algorithm
- (b) Elevator algorithm
- (c) Greedy algorithm
- (d) Banker's algorithm

**Answer: (b)**

SCAN sweeps the head in one direction then reverses, like an elevator.

**Q32**

DSSSB TGT

All 3 exams

**Spooling is most commonly associated with which device?**

- (a) Keyboard
- (b) Printer
- (c) Monitor
- (d) Mouse

**Answer: (b)**

Spooling (Simultaneous Peripheral Operation On-Line) queues jobs for slow devices like printers.

## SECTION B — Practice (full syllabus coverage for revision)

Organised by sub-topic.

### B1. Basics & types

**Q33**

All 3 exams

**Which is system software?**

- (a) MS Word
- (b) Operating system
- (c) Chrome
- (d) Photoshop

**Answer: (b)**

The OS is system software; the others are applications.

**Q34**

All 3 exams

**Multiprogramming aims to:**

- (a) Reduce memory
- (b) Keep the CPU busy by switching among jobs
- (c) Encrypt data
- (d) Increase disk size

**Answer: (b)**

Multiprogramming keeps several jobs in memory so the CPU switches to another when one waits for I/O.

**Q35**

All 3 exams

**A distributed OS:**

- (a) Runs on one CPU
- (b) Makes independent machines appear as one system
- (c) Has no kernel
- (d) Is a real-time OS

**Answer: (b)**

A distributed OS coordinates multiple machines to appear as a single system.

**Q36**

SSC/other

All 3 exams

**The command interpreter of an OS is the:**

- (a) Kernel
- (b) Shell
- (c) Driver
- (d) Loader

**Answer: (b)**

The shell interprets user commands and passes them to the kernel.

**Q37**

DSSSB TGT

All 3 exams

**A hard real-time system:**

- (a) Tolerates missed deadlines
- (b) Must never miss a deadline
- (c) Has no deadlines
- (d) Is a batch system

**Answer: (b)**

Hard RTOS deadlines are absolute; a miss is a failure (e.g. airbag).

## B2. Processes & threads

**Q38**

All 3 exams

**Threads of the same process share:**

- (a) Their own memory only
- (b) The process's code & data (but own stack)
- (c) Nothing
- (d) Separate PCBs entirely

**Answer: (b)**

Threads share the process's code/data/heap but have their own stack and registers.

**Q39**

All 3 exams

**Which is NOT a valid process state?**

- (a) Ready
- (b) Running
- (c) Waiting
- (d) Compiling

**Answer: (d)**

States: New, Ready, Running, Waiting, Terminated. 'Compiling' is not one.

**Q40**

All 3 exams

**The dispatcher's job is to:**

- (a) Pick the next process (scheduling)
- (b) Give the CPU to the selected process
- (c) Allocate memory
- (d) Handle page faults

**Answer: (b)**

The scheduler selects; the dispatcher actually hands the CPU to the chosen process (dispatch latency).

**Q41**

All 3 exams

**A zombie process is one that:**

- (a) Is running
- (b) Has finished but its entry remains in the process table
- (c) Is waiting for I/O
- (d) Never started

**Answer: (b)**

A zombie has terminated but its PCB entry lingers until the parent reads its exit status.

### B3. CPU scheduling

**Q42**

All 3 exams

**SRTF is the preemptive version of:**

- (a) FCFS
- (b) SJF
- (c) Round Robin
- (d) Priority

**Answer: (b)**

Shortest Remaining Time First = preemptive SJF.

**Q43**

All 3 exams

**Which metric is the time from submission to completion?**

- (a) Waiting time
- (b) Turnaround time
- (c) Response time
- (d) Burst time

**Answer: (b)**

Turnaround time = completion – arrival. Waiting = turnaround – burst.

**Q44**

All 3 exams

**A very small time quantum in Round Robin leads to:**

- (a) Convoy effect
- (b) High context-switch overhead
- (c) Starvation
- (d) Deadlock

**Answer: (b)**

Too-small quantum causes frequent context switches, wasting CPU on overhead.

**Q45**

GATE-level

Coal India MT

**Multilevel feedback queue scheduling allows a process to:**

- (a) Stay in one queue
- (b) Move between queues based on behaviour
- (c) Never be preempted
- (d) Skip the CPU

**Answer: (b)**

Processes move between priority queues based on CPU/IO behaviour — flexible but complex.

## B4. Synchronization

**Q46**

All 3 exams

**A race condition occurs when:**

- (a) One process runs alone
- (b) Output depends on the timing of concurrent access to shared data
- (c) Memory is full
- (d) A deadlock happens

**Answer: (b)**

A race condition = result depends on interleaving/timing of concurrent accesses to shared data.

**Q47**

All 3 exams

**A counting semaphore is used to manage:**

- (a) A single resource
- (b) Multiple identical resources
- (c) Only the CPU
- (d) Only memory

**Answer: (b)**

A counting semaphore tracks a pool of identical resources (value = available count).

**Q48**

All 3 exams

**Which classic problem illustrates synchronization with a bounded buffer?**

- (a) Dining Philosophers
- (b) Producer-Consumer
- (c) Readers-Writers
- (d) Sleeping Barber

**Answer: (b)**

Producer-Consumer (bounded buffer) is the canonical bounded-buffer synchronization problem.

**Q49**

All 3 exams

**A monitor differs from a semaphore in that it:**

- (a) Is lower level
- (b) Bundles shared data with the procedures guarding it
- (c) Cannot block
- (d) Is hardware

**Answer: (b)**

A monitor is a higher-level construct encapsulating shared data + its access procedures (with condition variables).

## B5. Deadlocks

**Q50**

All 3 exams

**The 'hold and wait' condition means a process:**

- (a) Holds no resources
- (b) Holds some resources while waiting for others
- (c) Releases all resources
- (d) Never waits

**Answer: (b)**

Hold-and-wait: a process holds allocated resources while requesting more.

**Q51**

All 3 exams

**Deadlock detection in single-instance resources uses a:**

- (a) Page table
- (b) Resource-allocation graph (cycle = deadlock)
- (c) Ready queue
- (d) TLB

**Answer: (b)**

A cycle in the resource-allocation graph indicates deadlock when resources are single-instance.

**Q52**

All 3 exams

**The 'Ostrich algorithm' for deadlocks means:**

- (a) Prevent them
- (b) Ignore the problem
- (c) Detect and recover
- (d) Avoid via Banker's

**Answer: (b)**

The Ostrich approach ignores deadlocks (used by many general-purpose OS as they're rare).

**Q53****GATE-level****Coal India MT**

**In Banker's Algorithm, a 'safe state' guarantees:**

- (a) No process ever waits
- (b) There exists an order to satisfy all processes without deadlock
- (c) Maximum CPU use
- (d) Zero page faults

**Answer: (b)**

A safe state means a safe sequence exists in which every process can obtain its max resources and finish.

## **B6. Memory management**

**Q54****All 3 exams**

**In paging, logical memory is divided into \_\_\_\_ and physical memory into \_\_\_\_:**

- (a) frames; pages
- (b) pages; frames
- (c) segments; pages
- (d) blocks; segments

**Answer: (b)**

Logical memory → pages; physical memory → frames; the page table maps pages to frames.

**Q55****All 3 exams**

**Virtual memory allows:**

- (a) Programs larger than physical RAM to run
- (b) Faster CPU
- (c) More cores
- (d) Bigger disk

**Answer: (a)**

Virtual memory runs processes partly on disk, so they can exceed physical RAM size.

**Q56****All 3 exams**

**The TLB is a cache for:**

- (a) Data blocks
- (b) Page-table entries (address translations)
- (c) Instructions
- (d) Disk sectors

**Answer: (b)**

The Translation Lookaside Buffer caches recent page-table entries to speed address translation.

**Q57****All 3 exams**

**LRU replaces the page that was:**

- (a) Loaded first
- (b) Least recently used
- (c) Most recently used
- (d) Largest

**Answer: (b)**

LRU evicts the least-recently-used page, approximating the optimal policy.

**Q58****GATE-level****Coal India MT**

With page size 4 KB, how many bits are used for the offset?

- (a) 10
- (b) 12
- (c) 16
- (d) 8

**Answer: (b)**

4 KB =  $2^{12}$  bytes → 12 offset bits. GATE-style paging arithmetic.

## B7. File systems & disk

**Q59****All 3 exams**

Linked allocation of files does NOT support efficient:

- (a) Sequential access
- (b) Random access
- (c) File growth
- (d) Small files

**Answer: (b)**

Linked allocation must follow pointers from the start, so random access is slow.

**Q60****All 3 exams**

A directory in a file system is used to:

- (a) Store raw data
- (b) Organize and track files
- (c) Schedule the CPU
- (d) Manage RAM

**Answer: (b)**

Directories organize files and map names to their storage locations.

**Q61****All 3 exams**

In C-SCAN, after reaching one end the head:

- (a) Reverses immediately
- (b) Jumps to the other end and continues one way
- (c) Stops
- (d) Serves randomly

**Answer: (b)**

C-SCAN services in one direction, then jumps back to the start — giving more uniform wait times.

**Q62****GATE-level****Coal India MT**

Head at 50; requests served FCFS in order 82,170,43,140. Total head movement is:

- (a) 208
- (b) 344
- (c) 299
- (d) 250

**Answer: (b)**

$|50-82|+|82-170|+|170-43|+|43-140| = 32+88+127+97 = 344$  cylinders. FCFS simply sums consecutive absolute moves in arrival order.

## Coverage & next

### ✓ What this bank covered

PYQ-style + Practice across all OS sub-topics: basics/types, processes & scheduling, synchronization & deadlocks, memory & virtual memory, file systems & disk.

GATE-tagged numericals (scheduling, page faults, Banker's, head-movement, paging bits) are Coal-India depth — skip for IBPS-only.

Note: original questions modelled on real exam patterns, not verbatim reproductions.